

de Broglie Wave phase and group velocities

1. **de Broglie wave:** The de Broglie wavelength λ for a particle that has certain mass (rest mass m_0) is related to its momentum p as

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v} \dots \dots \dots (1)$$

Where v is the velocity of the particle, and $\gamma = \frac{1}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$

If $v \ll c$ (velocity of light) then

$$\lambda = \frac{h}{p} = \frac{h}{m_0 v} \dots \dots \dots (2)$$

Since the de Broglie wave is associated with a moving body one can expect that the wave should have the same velocity as the particle.

Let us check?

If we assume v_p the velocity of the de Broglie wave associated with the particle and v is the velocity of the particle.

2. **Phase Velocity:** The velocity v_p with which the de Broglie wave travels is called phase velocity and is defined by following equation (3).

The according to wave theory we can write

$$v_p = v\lambda \dots \dots \dots (3)$$

Where v is the frequency of the de Broglie wave and λ is the wavelength of the de Broglie wave.

So according to de Broglie the wavelength λ can be written as

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v} \dots \dots \dots (4)$$

We also know that from mass energy relation

$$E = hv = \gamma m_0 c^2$$

$$v = \frac{\gamma m_0 c^2}{h} \dots \dots \dots (5)$$

Now from equation (3) we can find out phase velocity of the de Broglie wave using equations (4) and (5).

$$v_p = \frac{\gamma m_0 c^2}{h} \times \frac{h}{\gamma m_0 v}$$

$$v_p = \frac{c^2}{v} \dots \dots \dots (6)$$

From equation (6) we infer that velocity of the de Broglie wave is faster than the velocity of particle as $v \ll c$. Further since $v \ll c$ so $v_p \gg c$, which means de Broglie wave travels much faster than the velocity of light. This is wrong result as phase velocity cannot be greater the light velocity. ***In order to understand this discrepancy we must look the distinction between phase and group velocity.***

- Group velocity(v_g):** de Broglie wave associated with a moving particle consists of a group of wave rather a single wave and velocity with which this group of wave packet travel is called the group velocity.

$$\text{Group velocity } v_g = \frac{d\omega}{dk} \dots \dots \dots (7)$$

Let us check whether the group of wave associated with the moving particle travels with the same velocity as particle moves?

Since we know that the angular frequency ω is defined as

$$\omega = 2\pi\nu \dots \dots \dots (8)$$

Using equation (5)
$$\omega = \frac{2\pi\gamma m_0 c^2}{h} \dots \dots \dots (9)$$

$$\omega = \frac{2\pi m_0 c^2}{h \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{2}}} \dots \dots \dots (10)$$

Also wave number

$$k = \frac{2\pi}{\lambda} \dots\dots\dots(11)$$

Using equation (4) we can get

$$k = \frac{2\pi m_0 v}{h \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{2}}} \dots\dots\dots(12)$$

From equation (7) group velocity of the group of waves can be written as

$$v_g = \frac{d\omega}{dk} = \frac{d\omega/dv}{dk/dv} \dots\dots\dots(13)$$

After putting the values of $\frac{d\omega}{dv}$ and $\frac{dk}{dv}$ after differentiating equations (10) and (12)

With respect to v , we get

$$v_g = v \dots\dots\dots(14)$$

Thus the de Broglie wave group associated with the moving particles travel with the same velocity as the particle.